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A Novel Solution to Control Payzone Losses

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Abstract

Downhole losses is one of the most challenging and costly events in drilling and workover operations. When it comes to controlling losses in the reservoir section, it becomes more critical due to the special design needed for loss circulation material (LCM). These LCM should be reservoir friendly such that they are easily removable or do not create a source of permanent damage. This work investigates a novel solution of utilizing pre-crosslinked polymer gel systems to control downhole losses in brines without jeopardizing the reservoir quality.

Two different pre-crosslinked polymer gel systems were evaluated as fluid loss control for use with completion or workover brines in a 50:50 ratio. The systems underwent fluid loss control, dissolution efficiency and regain flow tests. Fluid loss control test was conducted on 10 micron ceramic disk by placing the mixed fluid in HPHT cell at different pressures up to 1,500 psi and recording the fluid loss volume at multiple time durations. The generated filtercake was immersed in acid treatments for 24 hours to assess the external filtercake solubility. Furthermore, regain flow on the ceramic disk prior to forming the filtercake and after the treatment was measured to evaluate both the remaining external and internal filtercake damage.

Based on the findings of this study, it was concluded that the pre-crosslinked polymer gel systems were very efficient in controlling the brine fluid loss by creating both external and internal filtercakes. Initial spurt loss of brines from the evaluated systems was about 15-25%, with differential pressures between 500 – 1,500 psi and temperatures of 150-250°F. In addition, external filtercake dissolution showed efficient solubility in acid treatment ranging from 85 to 91 wt.%. However, the internal filtercake of the used ceramic disk after remediation was not adequately removed. Thus, regain flow of 11-13% was observed using two test methods i.e. constant time with an applied pressure of 5 psi and constant volume with an applied pressure of 100 psi. Although the internal filtercake dissolution showed low solubility, the system can efficiently work in carbonate formations due to its reactivity with acid.

This work evaluates pre-crosslinked polymer gel systems to control downhole losses in completion and workover applications. It focuses on how friendly the system in minimizing losses efficiently and addressing the best methods to be applied in the field without affecting reservoir quality.

Introduction

Drilling fluids are fundamental to the efficiency and safety of drilling operations, serving multiple essential functions: controlling formation pressures, suspending and removing cuttings from the well, sealing permeable formations to minimize formation damage, and cooling and lubricating the drilling assembly (Caenn et al., 2011). Nevertheless, despite these crucial functions, drilling muds often encounter significant challenges such as fluid invasion, poor hole cleaning, lost circulation, and shale swelling. These issues can lead to increased operational costs, well control incidents, and permanent damage to the reservoir. Fluid invasion refers to the migration of filtrate into the formation, leading to blocked permeability in the pay zone, reduced hydrocarbon productivity, and significantly contribute to increased non-productive time (NPT) (Deville et al., 2022; Taheri & Torab, 2023). The risk of formation damage is especially acute in sensitive reservoirs, where even minimal invasion can have lasting negative impacts. Consequently, maintaining formation integrity and minimizing formation damage are of paramount importance. The severity of formation damage grows particularly when drilling across sensitive reservoir formations. Among the most persistent and costly challenges faced during drilling and completion operations is fluid loss, which can directly impact hydrocarbon recovery and well integrity (Nguyen et al., 1996).

To address these challenges, drill-in fluids (DIFs) are specifically designed for use in reservoir intervals, with a focus on reducing formation damage and maximizing productivity. Unlike conventional drilling muds, these fluids are specially formulated to exhibit certain characteristics: they should contain minimal solids, be acid-soluble for easy cleanup, and remain compatible with formation fluids (Ezell et al., 2010). In response to formation damage and fluid loss challenges, the industry has developed several fluid additives and engineered formulations. Conventional loss prevention techniques, such as increasing fluid viscosity or adding bridging materials (Scheuerman, 1983; Hudson et al., 1983), often fall short in fragile reservoir sections, where the risk of formation damage is high. Sized bridging particles, such as calcium carbonate (CaCO_3), may remain in the formation and may not be effectively removed during cleanup, leading to permanent impairment of well productivity (Scheuerman, 1983; Chang et al., 1998). To mitigate this issue, water-soluble polymers such as hydroxyethyl Cellulose (HEC) have been widely used to increase the viscosity of DIFs and enhance their ability to plug pore throats effectively. HEC is valued for its compatibility with water-based drilling fluids, biodegradability, and minimal toxicity (Songire & Uppuluri, 2014). However, its application in high-permeability and/or high temperature wells presents several limitations. Primarily, HEC provides insufficient filtration control properties and suboptimal temperature stability (Blauch et al., 1989). Additionally, due to their linear molecular structure, these polymers exhibit limited crosslinking potential under downhole conditions, which reduce their effectiveness in controlling severe fluid losses or in formations with varying permeability (Blauch et al., 1989). Consequently, linear gel systems are not suitable for use in high-permeability reservoirs (≥ 1000 md), high overbalance pressures, or elevated temperature environments, as they tend to deeply penetrate the formation and may cause lasting permeability damage (Cole et al., 1995; Songire & Uppuluri, 2014). The use of HEC can result in viscous plugging, the formation of fish eyes and microgels, and precipitation at high temperatures, leaving behind a solution with a significant amount of insoluble polymers (Hodge, 1998).

To overcome these limitations, the industry has developed various fluid loss control pills, which are generally categorized into three types: crosslinked polymers, bridging pills, and the more recent pre-crosslinked polymer gel systems (Vollmer & Lejeune, 2005). While non-crosslinked HEC and particulate-based fluid loss materials have shown potential for formation damage, bridging pills containing solids may still plug pores and be challenging to clean out from perforated zones (Blauch et al., 1989). In contrast, pre-crosslinked gel pills have proven to be more effective in high-permeability formations due to their solid-free and high-crosslinking nature (Vollmer & Lejeune, 2005). Given their crosslinking properties, minimal damage is expected, as they reduce the risk of deep penetration into the formation (Cole et al., 1995; Dewprashad et al., 1998; Gibson et al., 2011). First-generation crosslinked fluid loss control (FLC)

pills have proven effective in minimizing fluid loss compared to linear gels, but their on-site preparation poses operational and safety challenges. The process requires mixing and hydrating a crosslinkable polymer in brine, often with acid to lower pH, which is time-intensive and introduces health and environmental risks. Crosslinking is initiated just before pumping the pill downhole, aiming for it to set at the target zone. However, this in-situ crosslinking approach lacks control, risking premature gelation that can damage equipment or delayed crosslinking that may cause formation damage due to polymer invasion (Gibson et al., 2011). Advanced pre-crosslinked gel systems have addressed these challenges by eliminating the need for on-site mixing and in-situ crosslinking. These systems are pre-formulated and fully hydrated, allowing for immediate deployment at the rig, which reduces non-productive time and removes the risks associated with acid handling and uncontrolled crosslinking. As a result, they offer safer, more efficient, and more reliable fluid loss control, particularly in challenging wellbore conditions (Gibson et al., 2011).

Given the detrimental effects of uncontrolled fluid loss in reservoir sections, it is imperative to employ effective fluid loss management strategies, including customized DIF design, maintaining slightly overbalanced pressure, controlling fluid invasion into the formation, and incorporating real-time monitoring to prevent and remediate fluid loss. Pre-crosslinked gel polymer systems can be added directly to the completion brine, represent a novel and increasingly successful approach to overcoming these challenges. These robust systems combine the advantages of controlled gelation, improved thermal properties, ready-to-use systems, reliable performance, and reduced risk of formation damage, marking a significant advancement in the management of fluid loss during drilling and completion operations. This paper presents a thorough evaluation of two pre-crosslinked polymer gel systems L & H, specifically focusing on the cleanup efficiency of the fluid loss control systems in acidic brines, their fluid loss performance in different brines under different pressures, and the regain flow after filter cake treatment with acidic brine.

Experimental Procedures

Fluid Loss Control

To assess the ability of pre-crosslinked polymer gel systems to control fluid loss at simulated downhole conditions. The experiment was carried out by measuring the weight of a dry and saturated 10-micron ceramic disk, then placing it in an HPHT cell filled with a polymer gel system. The cell is pressurized to different levels (500, 1000, and 1500 psi), heated, and the fluid loss volume is recorded at various time intervals over 24 hours. After the test, the cell is cooled and depressurized, and the disk is removed and photographed for analysis.

Pre-Crosslinked Polymer Gel Systems Filtercake Dissolution Efficiency

The objective of this test is to evaluate the solubility of a treatment fluid on the pre-crosslinked polymer gel systems. The experiment consists of measuring the weight of a dry and saturated 10-micron ceramic disk, then placing it in an HPHT cell filled with a system. The cell is pressurized to different levels (500, 1000, and 1500 psi), heated, and the fluid loss volume is recorded at various time intervals over 24 hours. After the test, the cell is cooled and depressurized, and the disk is removed and photographed. The weight of the resulting filtercake is then measured, and it is immersed in an acid system for 24-48 hours to allow for reaction. The filtercake is then rinsed with DI water, and its final weight is measured to determine the percentage of dissolution.

Formation Damage Remediation

To test the efficacy of acidic brine to remove the pre-crosslinked polymer gel systems external and internal filtercake. The experiment comprises a series of steps to evaluate the performance of a pre-crosslinked polymer gel system. Initially, a dry 10-micron ceramic disk is weighed, vacuum saturated with DI water, and re-weighed to determine its saturation level. The disk is then placed in an HPHT cell, filled with DI

water, and subjected to a low pressure of 5 psi to measure the initial fluid throughput. This process is repeated to ensure consistent results. Next, the cell is filled with the pre-crosslinked polymer gel system, and a differential pressure of 500 psi is applied at 250°F. The fluid loss volume is recorded at various time intervals over 24 hours. After the test, the cell is cooled, depressurized, and emptied, and the disk with the resulting filtercake is carefully removed and submerged in an acid system for 24-48 hours. The filtercake is then rinsed, weighed, and photographed. Finally, the disk is re-installed in the HPHT cell, filled with DI water, and subjected to the same low-pressure test as before to determine the percentage of return flow.

Fluid Loss Control Results

Tables 1 and 2 show the fluid loss test results for system L in NaCl brine and System H in CaCl₂ brine at 250°F with differential pressures from 500 to 1,500 psi.

Table 1—System L in NaCl Brine Filtrate @ 250°F for 24 hours

Time, hrs	$\sqrt{\text{Time, hrs}}$	Filtrate Vol @ 500 psi, cc	Filtrate Vol @ 1,000 psi, cc	Filtrate Vol @ 1,500 psi, cc
0	0.000	0	0	0
0.02	0.129	30	48	49
0.08	0.289	34	50	52
0.50	0.707	42	56	58
1.00	1.000	48	60	61
4.00	2.000	67	67	69
24.00	4.899	131	140	140

Table 2—System H in CaCl₂ Brine Filtrate @ 250°F for 24 hours

Time, hrs	$\sqrt{\text{Time, hrs}}$	Filtrate Vol @ 500 psi, cc	Filtrate Vol @ 1,000 psi, cc	Filtrate Vol @ 1,500 psi, cc
0	0.000	0	0	0
0.02	0.129	33	36	38
0.08	0.289	35	38	39
0.50	0.707	41	45	49
1.00	1.000	46	48	52
4.00	2.000	66	66	80
24.00	4.899	123.5	120	130

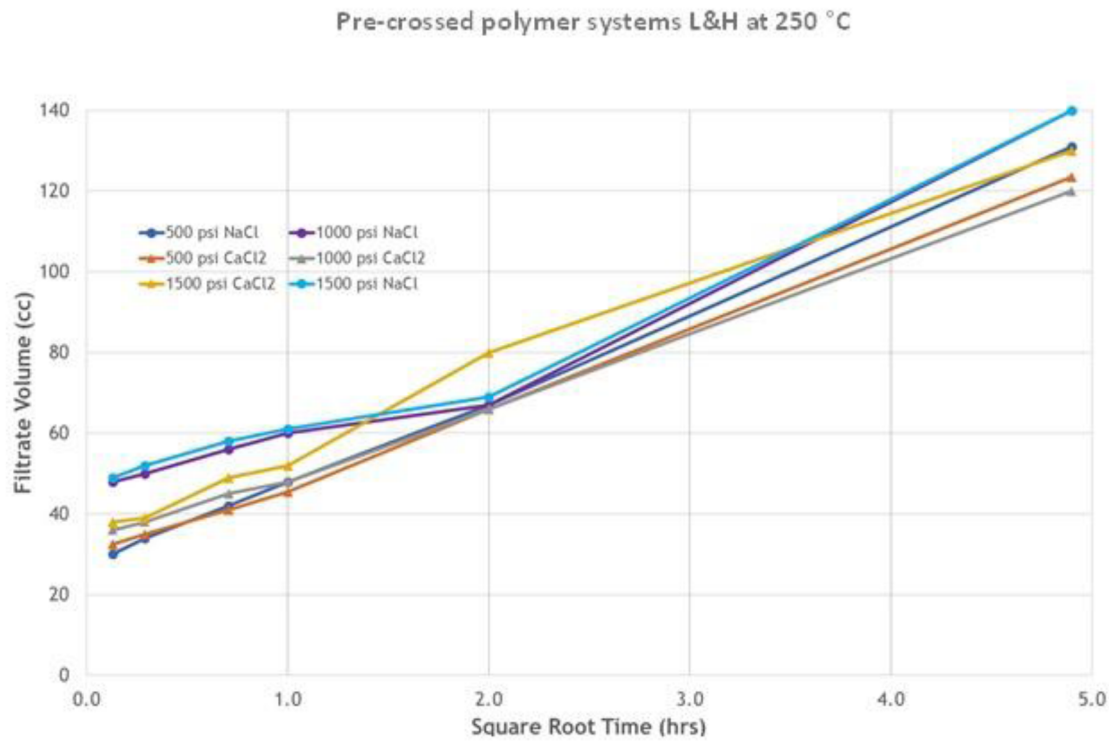


Figure 1—Fluid Loss for System L & H in NaCl and CaCl2 Brines @ 250°F

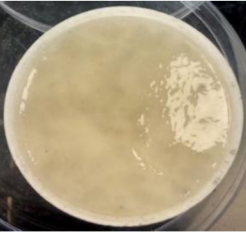
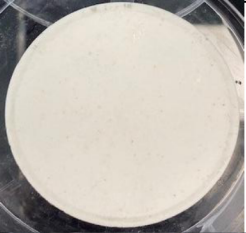
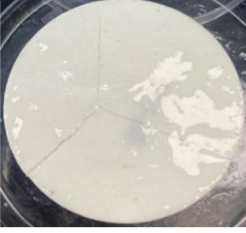
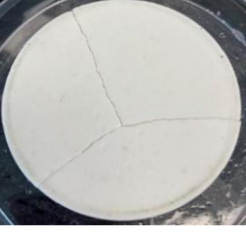
Filtercake Dissolution Efficiency

Tables 3 and 4 show the filtercake solubility test results after exposure to the recommended acidic brine systems. Overall, the acid systems were able to effectively dissolve 85-91 wt% of the systems filtercake.

Table 3—Filtercake Dissolution Efficiency of 10 wt% Formic / 10 wt% Acetic Acidic Brine

Pre-Crossed Polymer System	Photo Before Acid	Photo After Acid	% Solubility
System L in 74 pcf NaCl Brine			91%
System H in 85 pcf CaCl2 Brine			88%

Table 4—Filtercake Dissolution Efficiency of 5 wt% HCl / 10 wt% Acetic Acidic Brine

Pre-Crossed Polymer System	Photo Before Acid	Photo After Acid	% Solubility
System L in 74 pcf NaCl Brine			85%
System L in 85 pcf CaCl ₂ Brine			89%
System H in 85 pcf CaCl ₂ Brine			88%

Formation Damage Remediation

The regain flow after filtercake treatment with acidic brine was investigated by injecting de-ionized water through 10 micron ceramic disks either for a specified time or a fixed volume. As shown in Table 5 below, the test with an applied pressure of 5 psi for 20 seconds resulted in a final water injectivity of only 3 ml compared to initial injectivity of 23 ml (regain flow of 13%). This slower injectivity is primarily attributed to the internal filtercake since majority of the visible external filtercake was dissolved.

Table 5—Regain Flow Efficiency after 5 wt% HCl / 10 wt% Acetic Treatment on System L

Regain Injectivity Test	Flow Results @ 5 psi (ml of DI Water/20 sec)
Initial injectivity	23
Final injectivity	3

The applied pressure and the volume of water flow through the ceramic disk was increased to 100 psi and 300 ml, respectively, to see if this may improve the systems cleanup characteristics. Table 6 below, shows that the final injectivity of water took about 9 times longer to flow through compared to the initial rate (or a regain flow of 11%). This slower injectivity is again attributed to the internal filtercake since majority of the visible external filtercake was dissolved.

Table 6—Regain Flow Efficiency after 10 wt% Formic/10 wt% Acetic Treatment on system H

Regain Injectivity Test	Flow Results @ 100 psi (sec/300 ml of DI Water)
Initial injectivity	28
Final injectivity	264

Conclusions

The pre-crosslinked polymer gel systems L and H were evaluated as fluid-loss control for brines during completion and workover operations. The study scheme included; brine compatibility, fluid loss control, systems filtercake dissolution efficiency, and formation damage remediation. Based on this study, the followings were concluded:

- The systems can be prepared in NaCl, CaCl₂ and CaCl₂/CaBr₂ completion brines
- The systems are fluid loss control system for with wall building characteristics – resulting in both an external and internal polymer-based filtercake
- Initial spurt loss of brine from the system is about 15-25%, with differential pressures between 500-1,500 psi and temperatures of 150-250°F
- Systems external filtercake solubility in the recommended acidic brine systems ranged from 85 to 91 wt%
- Remediation tests with acidic brine on filtercake resulted in damaged ceramic disks with regain flow of 11-13% using two test methods – constant time with an applied pressure of 5 psi and constant volume with an applied pressure of 100 psi
- In these laboratory experiments, system's internal filtercake containing long-chain gel molecules remained trapped by the pore structure even after acid soaking

In conclusion, the proposed fluid loss control system demonstrated adequate efficacy in controlling/reducing brine leak-off with applied overbalance pressures up to 1500 psi and 250°F. Despite the mixed results of the polymeric residue clean-up with acidic brine (good external filtercake removal vs poor internal filtercake removal) in the laboratory tests on ceramic disks, its field use in carbonate reservoirs would be expected to be much better. Therefore, systems L and H are only recommended for carbonate reservoirs.

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